Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Match the Graph: Motion Sensor Lab

Unit 2: Forces and Motion

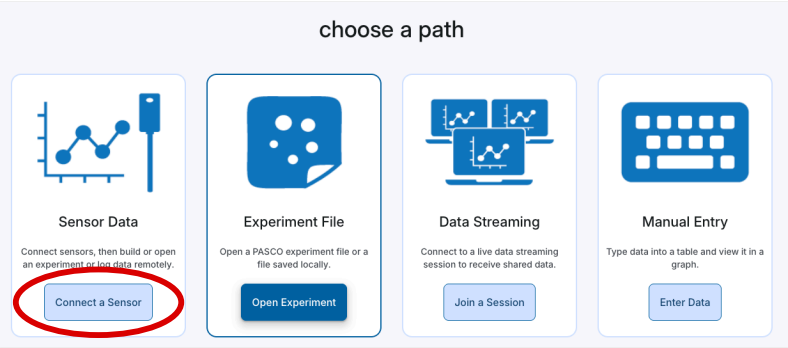
1. Open SPARKvue

Take out your **laptop**, open **Google Chrome**, and go to:

`sparkvue.pasco.com`

2. Connect the sensor

1. Put the sensor on a table, its **round screen at waist height**, facing **3–4 m** of clear floor.
2. Press the **power button**. The light **blinks red**.
3. In SPARKvue, on the **choose a path** screen, click **Connect a Sensor** (ringed), then pick the **number printed on your sensor's label**.

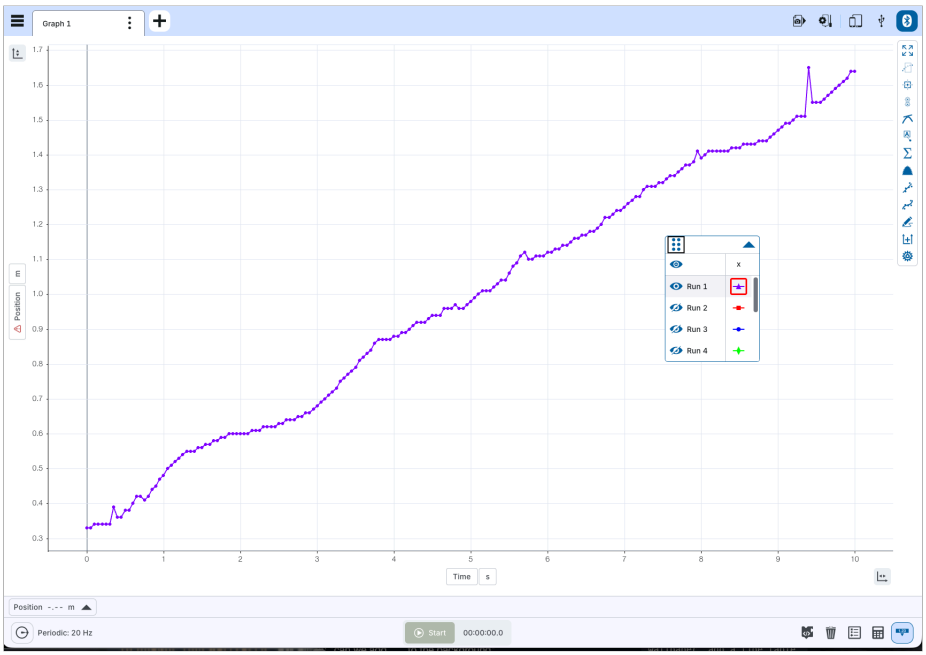


Check: the light now **blinks green**.

3. Show your position

1. In the list of measurements, click **Position (m)**.
2. Click **Graph**.

Check: a graph appears with **Position (m)** up the side and **Time (s)** along the bottom. It **rescales itself** to fit your walk, so read the numbers on *your* axes.

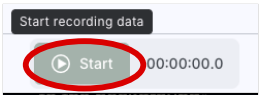


A position graph after a walk.

4. Walk a graph

The sensor measures **your distance from its round screen**. Walk **away** and the line goes **up**; walk **towards** it and the line goes **down**; stand still and it stays **flat**. Stay at least **0.5 m** away.

1. **Plan.** Pick a graph from the next pages. Read where its line starts and decide where to stand and how to move. Use the numbers: a line starting at 1.5 m means you start 1.5 m from the sensor.
2. **Stand** at the starting distance, facing the sensor.
3. **Walk it.** Your partner clicks **Start** (*Start recording data*), bottom of the screen. Move the way you planned, then your partner clicks **Stop** after **8 s**.



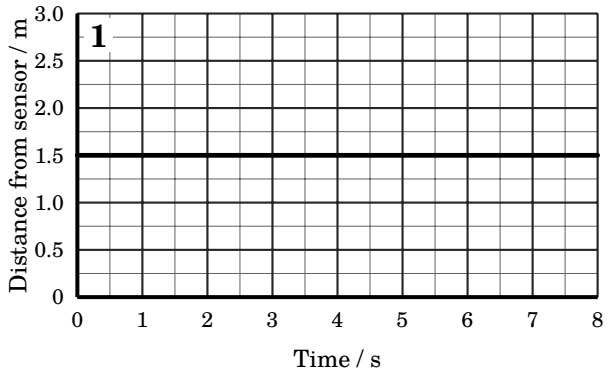
4. **Compare.** Read the numbers on your axes and hold your line against the graph on the sheet: same start distance, same steepness, same turns? Adjust the plan and walk it again until they match.
5. **Write it down.** Under the graph, describe how you moved, using the words in the box. Tick *Matched it?* when your line fits.

Check: everyone walks at least two graphs. **Care:** keep the floor in front of the sensor clear, and walk.

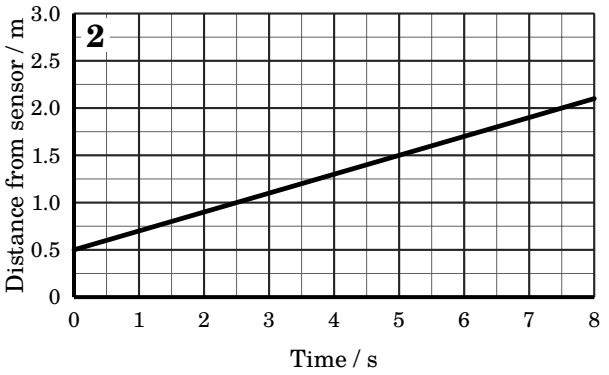
Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down

The six graphs

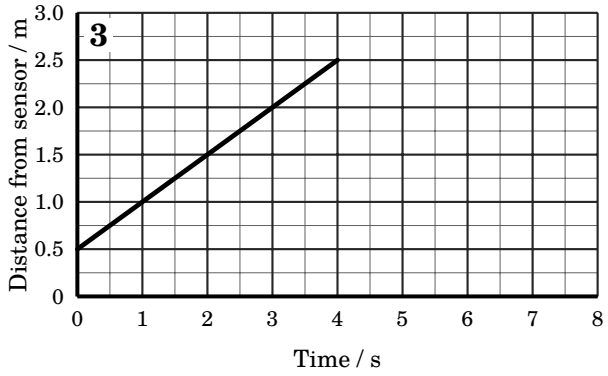
Work through them in order. Number 5 has four parts — describe each one.



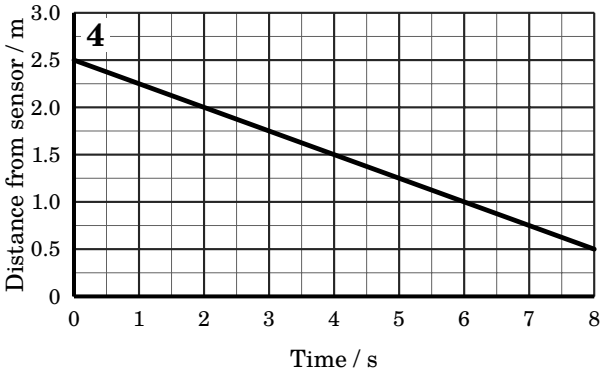
How I moved: **Matched it?** ☐



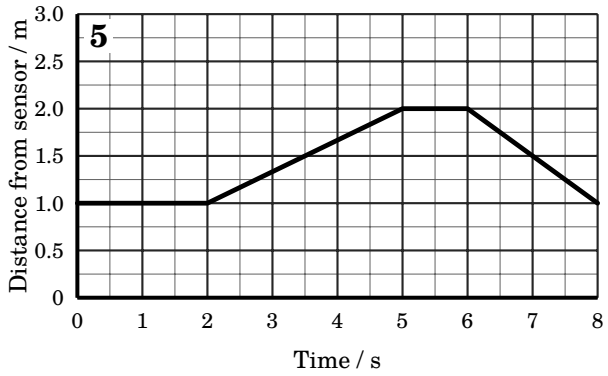
How I moved: **Matched it?** ☐



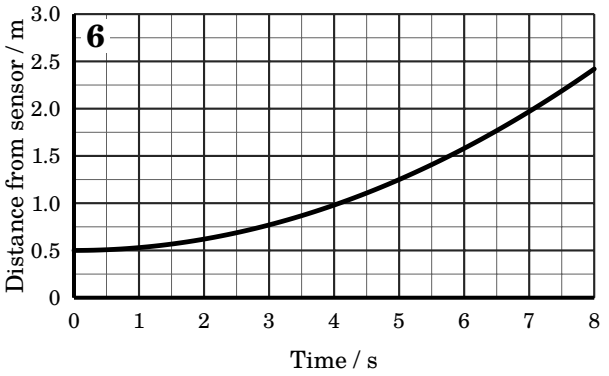
How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐

What the graphs show

1. On a distance–time graph, a flat line means the walker is _____.

2. Graphs 2 and 3 are both straight lines going up. Which one was the faster walk, and how can you tell from the graph?

_____

_____

3. Use Graph 3 to work out the walker's speed. They moved from 0.5 m to 2.5 m in 4 s.

speed = distance ÷ time = _____ m ÷ _____ s = _____ m/s

Check: the distance is how far they *moved*, end minus start.

To go further — the impossible graph

Nobody in the room can walk a graph that goes **straight up**, a vertical line. Explain why, using what a vertical line would mean about distance and time.

_____

_____

_____